

TuniCars+

Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

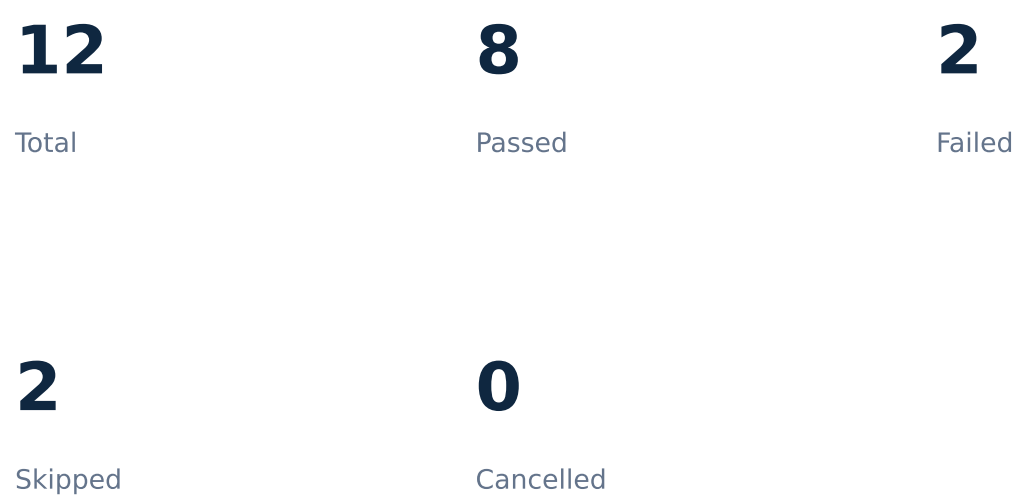
20260813_225853_1457cb

Date

2026-08-13T21:58:53.727367+00:00

Executive Summary

20260813_225853_1457cb



Overall measured availability: 55.05%

During this campaign, 8 of 12 scenarios passed, 2 failed and 2 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	kubernetes	swarm
container-kill	1	1
cpu	1	1
latency	1	1
memory	1	1
network-partition	1	1
node-failure	1	1

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
container-kill	swarm	1.812	11.047	True	True	9.05	17.796	10	9.09	PASS	N/A	N/A
node-failure	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	N/A Single-node Docker Swarm cannot demonstrate real node	N/A
cpu	swarm	N/A	N/A	N/A	Not Required	11.828	18.782	N/A	100.0	PASS	N/A	N/A
memory	swarm	N/A	N/A	N/A	Not Required	12.8	19.223	N/A	100.0	PASS	N/A	N/A
network-partition	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	FAIL	N/A	N/A Resilience-test authentication failed with HTTP
latency	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	FAIL	N/A	N/A Resilience-test authentication failed with HTTP 500
container-kill	kubernetes	6.109	17.609	True	True	1365.69	2052.195	7	22.22	PASS	N/A	N/A
node-failure	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	N/A Kubernetes inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL	N/A
cpu	kubernetes	N/A	N/A	N/A	Not Required	49.345	174.324	N/A	100.0	PASS	N/A	N/A
memory	kubernetes	N/A	N/A	N/A	Not Required	44.386	193.017	N/A	100.0	PASS	N/A	N/A
network-partition	kubernetes	0.156	0.031	True	True	36.7	112.733	15	N/A	PASS	N/A	N/A
latency	kubernetes	4.562	5.031	True	True	4522.318	4533.362	10	9.09	PASS	N/A	N/A

swarm — container-kill

State: **Pass**
Detection Time (s): 1.812

Recovery Time (s): 11.047

HTTP Success: N/A

HTTP Failures: 10

Availability (%): 9.09

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — node-failure

State: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Single-node Docker Swarm cannot demonstrate real node rescheduling.

swarm — cpu

State: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — memory

State: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — network-partition

State: Fail
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Resilience-test authentication failed with HTTP 500.

swarm — latency

State: Fail
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Resilience-test authentication failed with HTTP 500.

kubernetes — container-kill

State: **Pass**
Detection Time (s): 6.109
Recovery Time (s): 17.609

HTTP Success: N/A

HTTP Failures: 7

Availability (%): 22.22

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — node-failure

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

kubernetes — cpu

Status: Pass
Detection Time (s): N/A
Recovery Time (s): N/A
HTTP Success: N/A
HTTP Failures: N/A
Availability (%): 100.0
CPU (%): N/A
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — memory

Status: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — network-partition

Status: Pass
Detection Time (s): 0.156

Recovery Time (s): 0.031

HTTP Success: N/A

HTTP Failures: 15

Availability (%): N/A

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — latency

Status: Pass
Detection Time (s): 4.562

Recovery Time (s): 5.031

HTTP Success: N/A

HTTP Failures: 10

Availability (%): 9.09

CPU (%): N/A

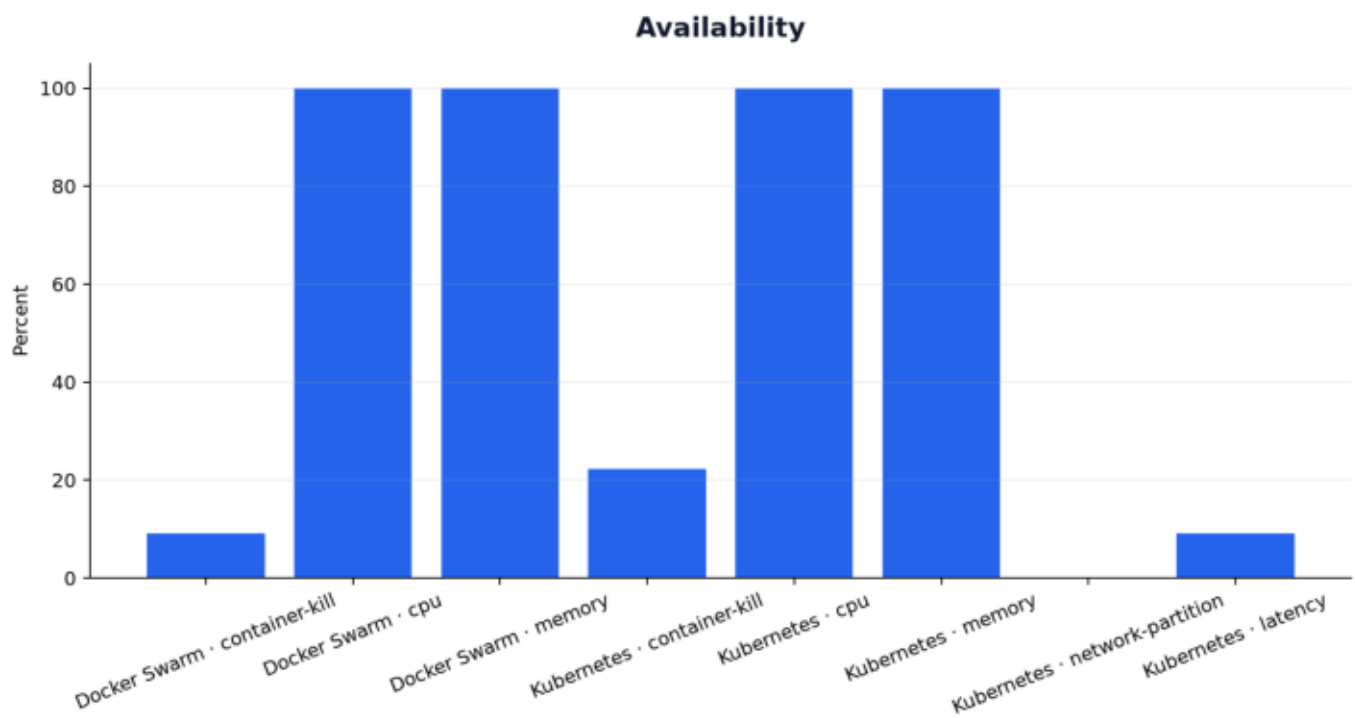
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

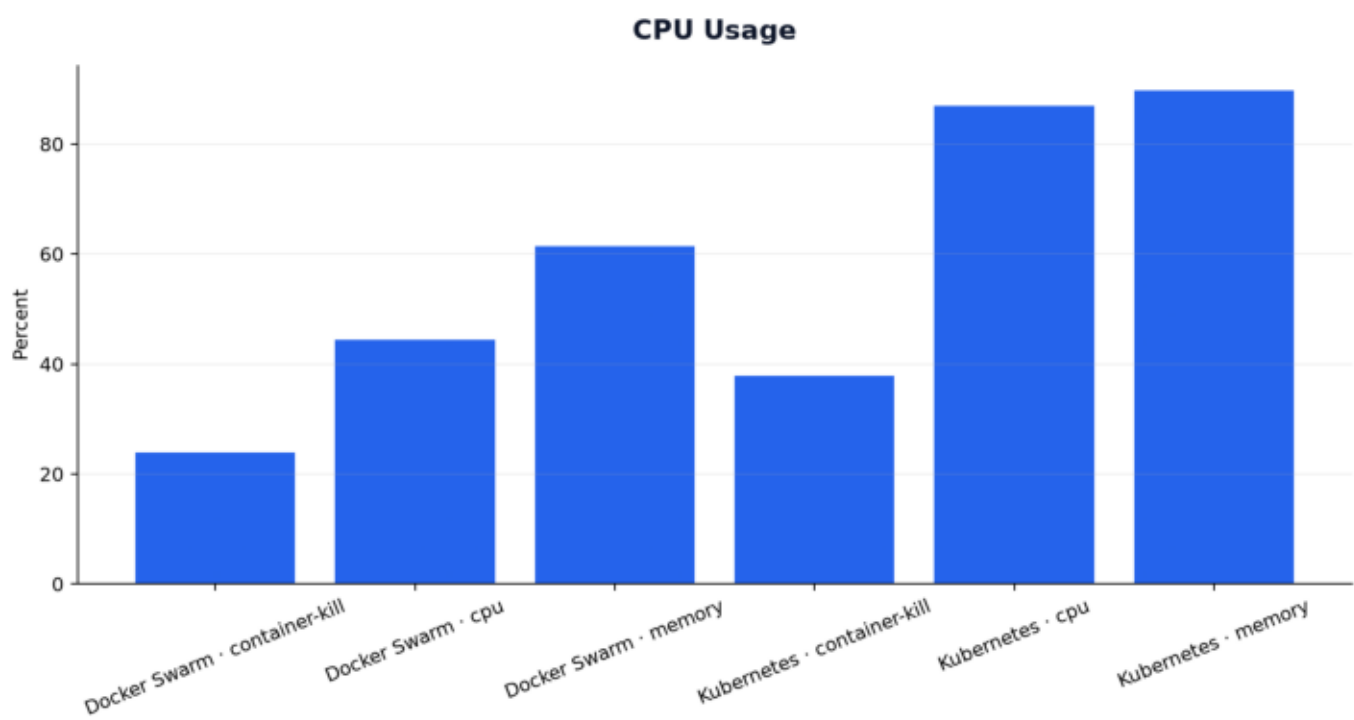
Availability

Current-session measured data



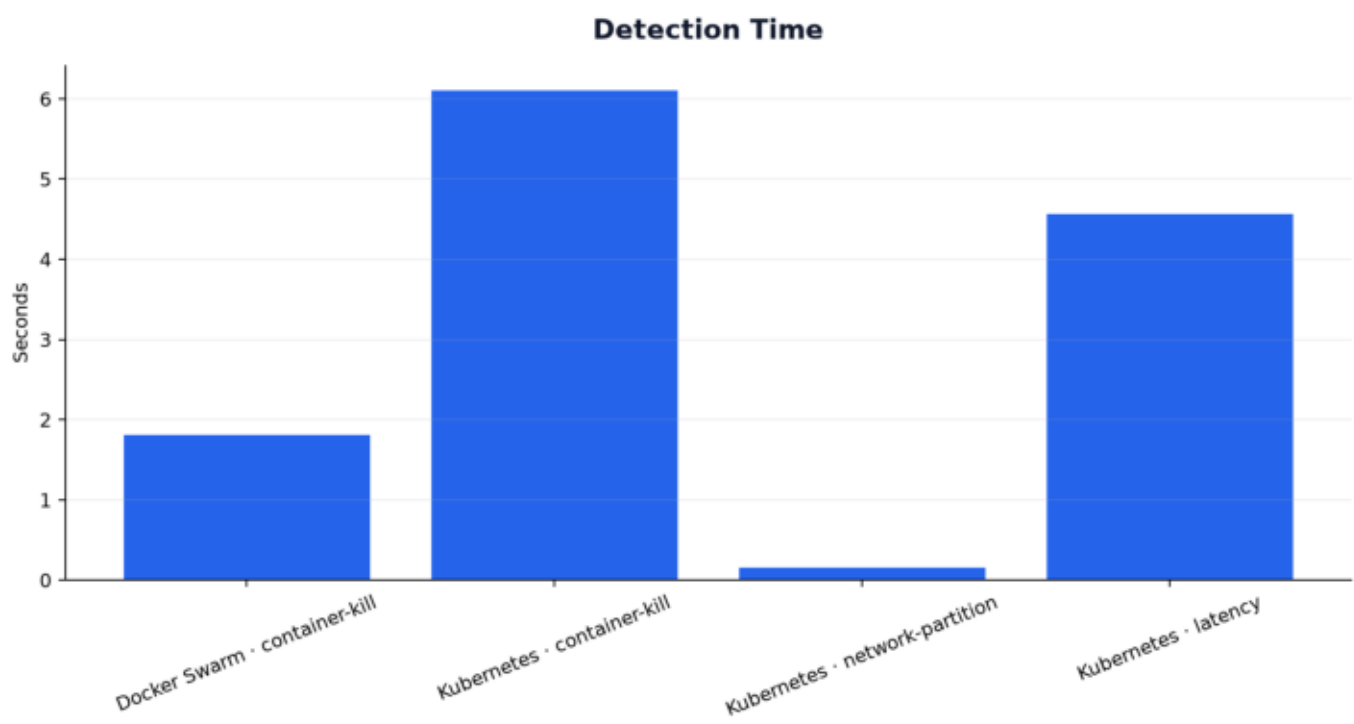
Cpu Usage

Current-session measured data



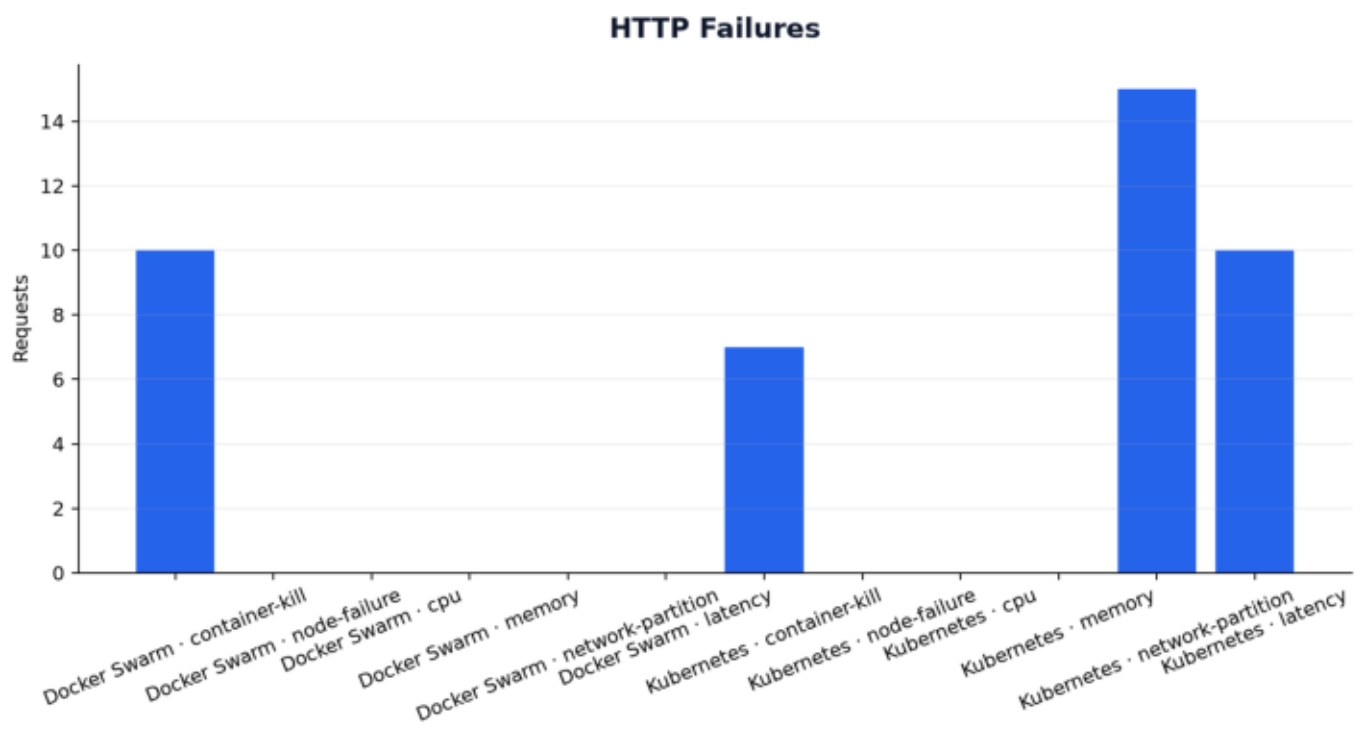
Detection Time

Current-session measured data



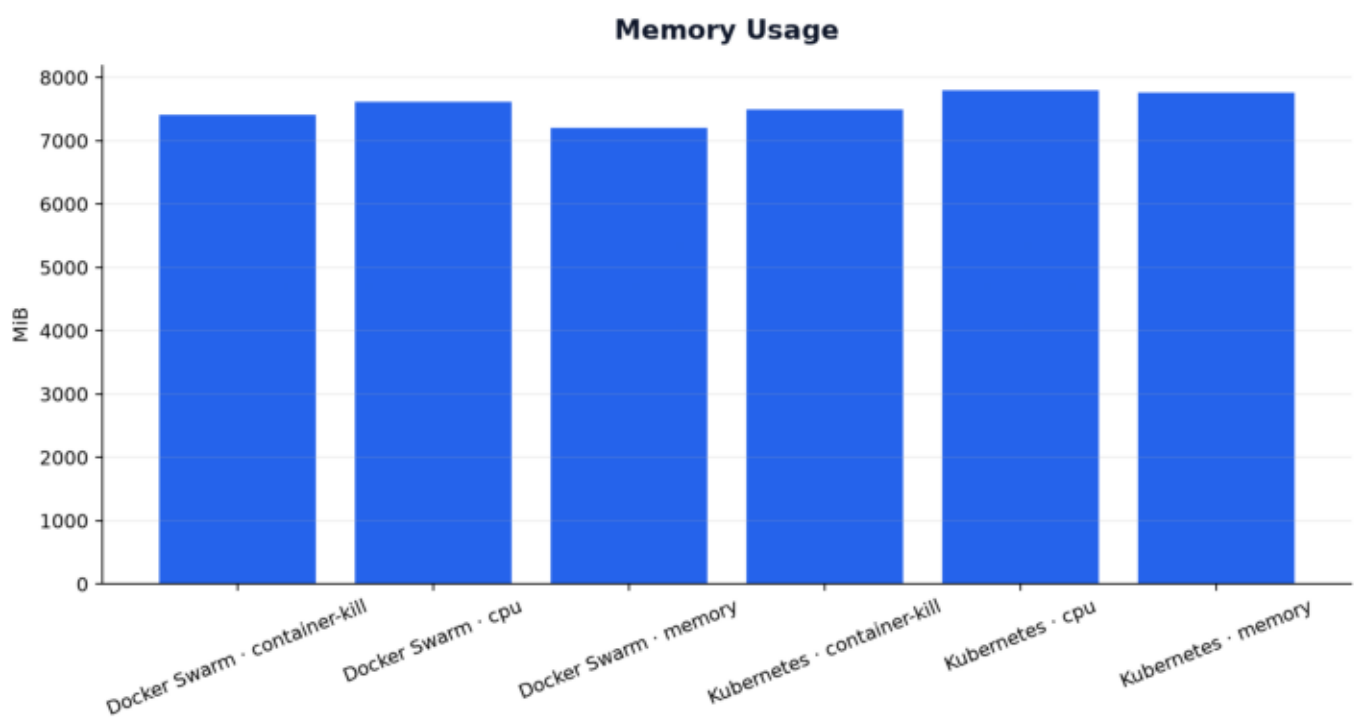
Http Failures

Current-session measured data



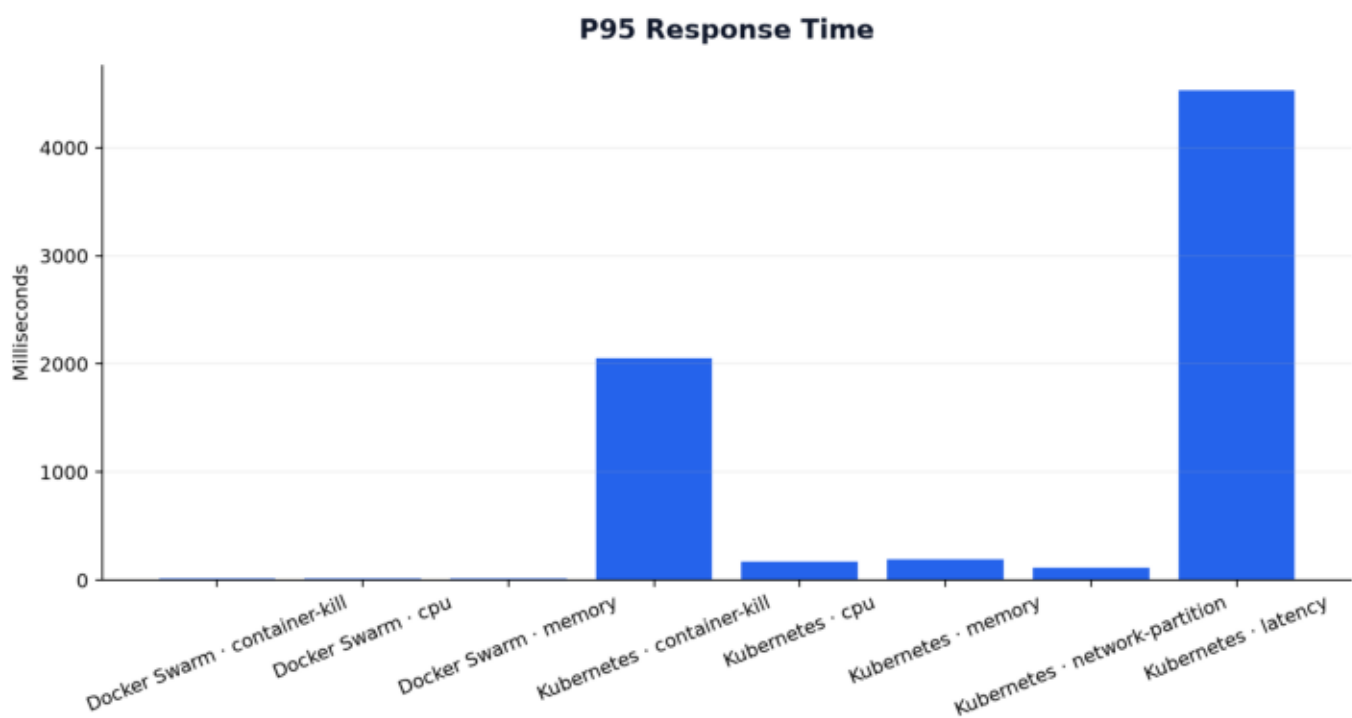
Memory Usage

Current-session measured data



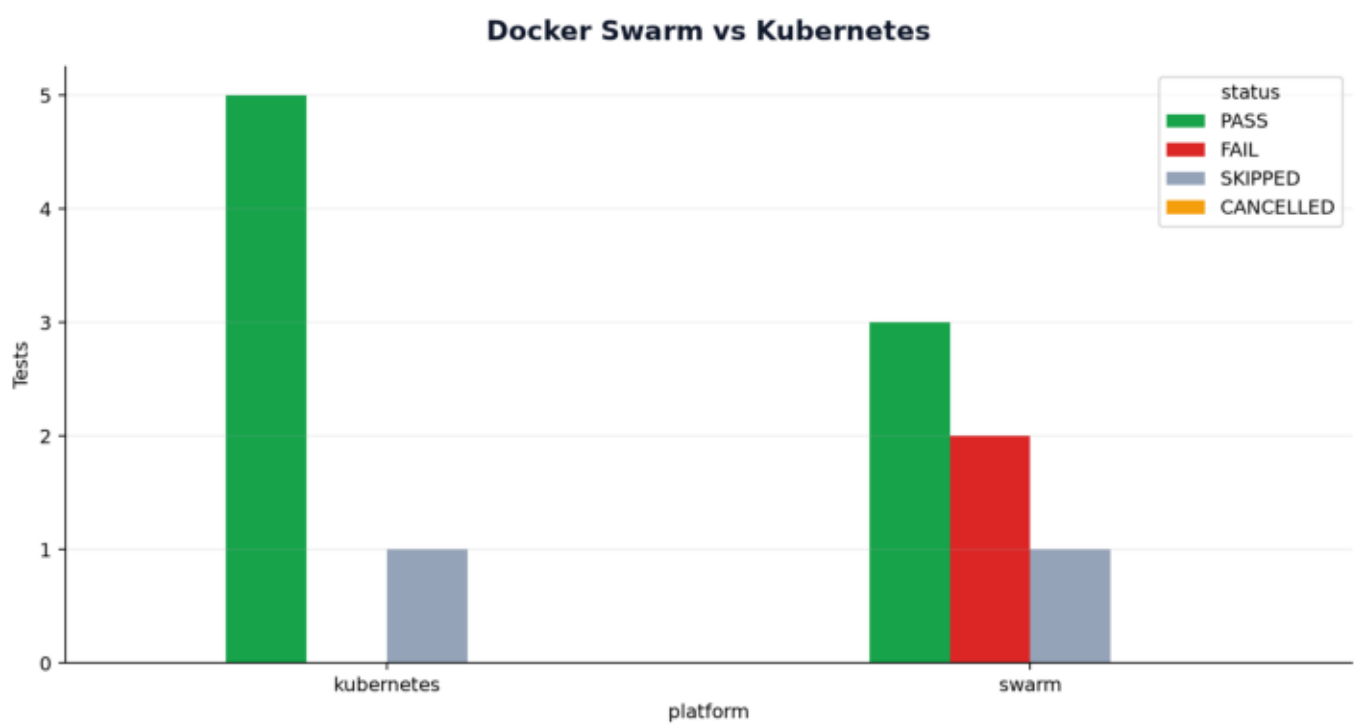
P95 Response Time

Current-session measured data



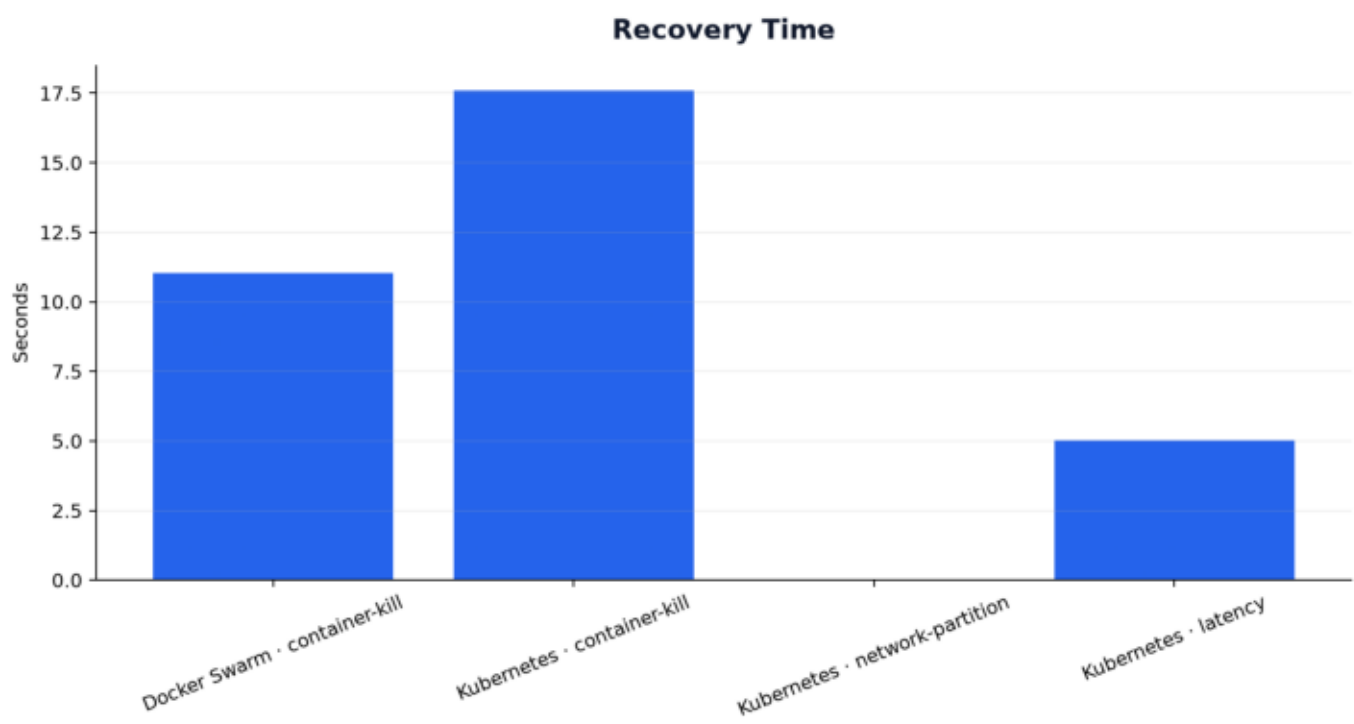
Platform Comparison

Current-session measured data



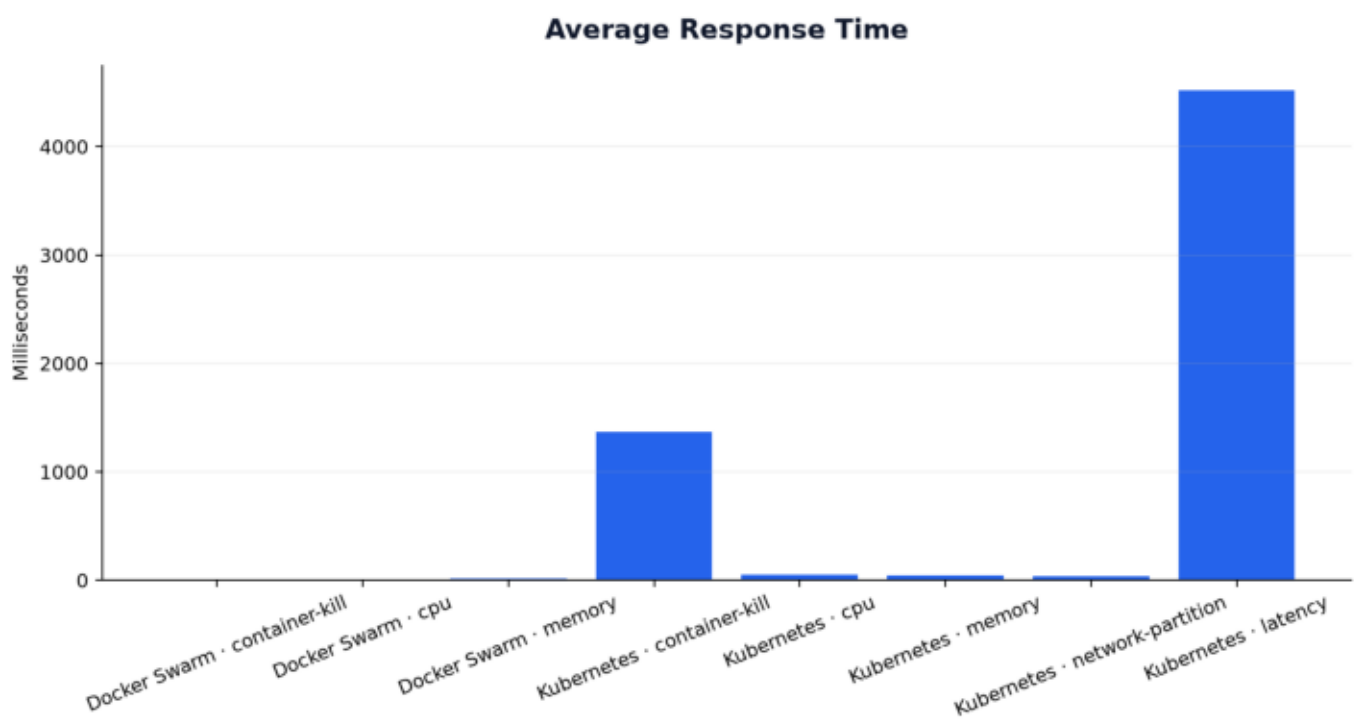
Recovery Time

Current-session measured data



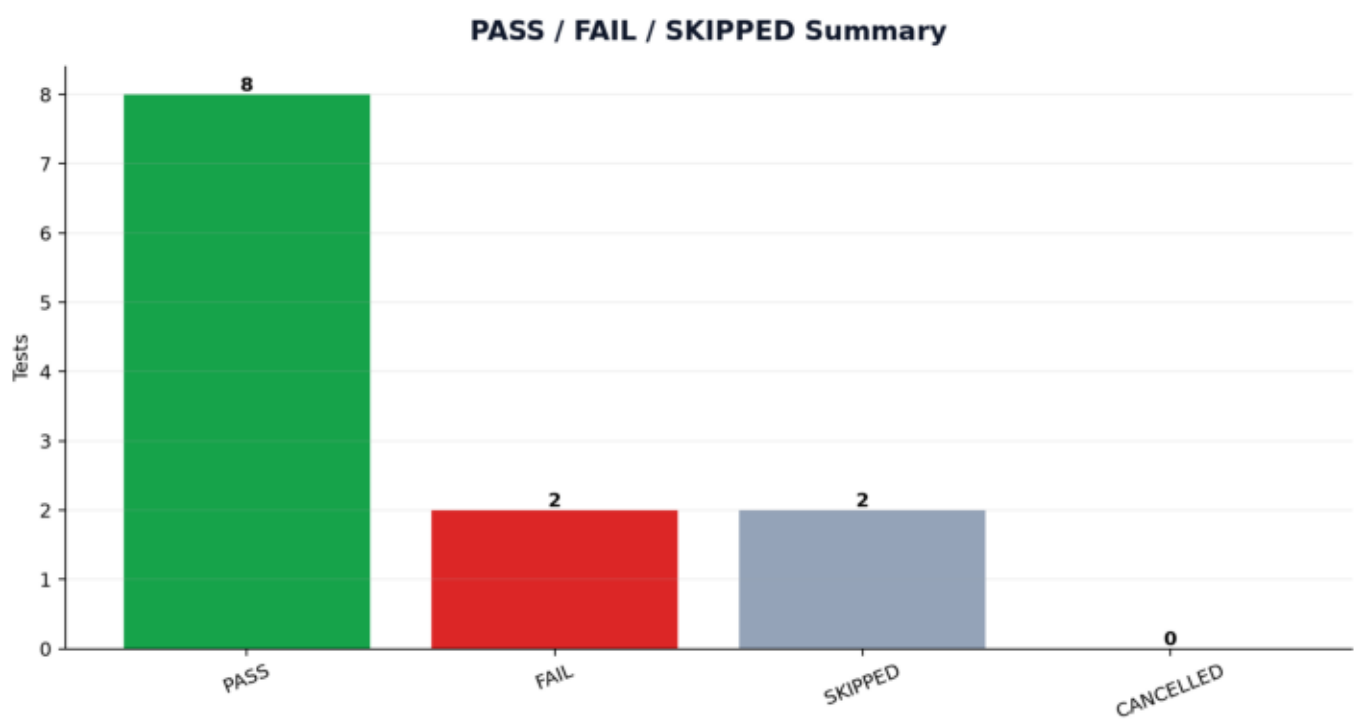
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
swarm	network-partition	FAIL	Resilience-test authentication failed HTTP 500.
swarm	latency	FAIL	Resilience-test authentication failed HTTP 500.

Skipped Tests

Platform	Scenario	Exact reason
swarm	node-failure	Single-node Docker Swarm cannot demonstrate real node rescheduling.
kubernetes	node-failure	Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

Swarm vs Kubernetes Comparison

Platform	FAIL	PASS	SKIPPED
kubernetes	0	5	1
swarm	2	3	1

Final Conclusion

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During this campaign, 8 of 12 scenarios passed, 2 failed and 2 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.